

UNIVERSITY OF TRANSPORT HO CHI MINH CITY

INFORMATION SECURITY

OVERVIEW OF CRYPTOGRAPHY AND DATA HIDING TECHNIQUES

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OVERVIEW OF CRYPTOGRAPHY

ORIGINS OF CRYPTOGRAPHY

History of Cryptography

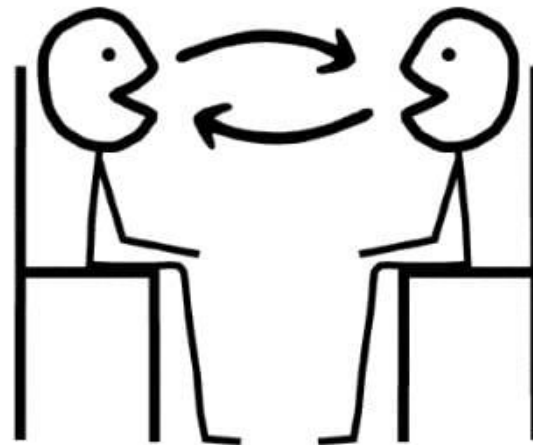
Human beings have long possessed two fundamental needs:

- Communication and information sharing
- Selective communication

These two needs led to the art of encoding messages in such a way that only the intended recipients could access the information. Unauthorized individuals would be unable to extract any meaningful information, even if the scrambled message fell into their hands.

The art and science of concealing messages to achieve confidentiality in information security is known as **cryptography**.

The term “**cryptography**” is derived from two Greek words: “**Kryptos**” meaning *hidden*, and “**graphein**” meaning *writing*.



ORIGINS OF CRYPTOGRAPHY

History of Cryptography

The art of cryptography is considered to have emerged alongside the development of writing itself. As civilizations evolved, people became organized into tribes, groups, and kingdoms. This led to the emergence of concepts such as power, warfare, supremacy, and politics.

These developments further intensified the natural human need for secret communication with selected recipients, thereby ensuring the continuous evolution of cryptography.



The origins of cryptography can be traced back to ancient Roman and Egyptian civilizations.

ORIGINS OF CRYPTOGRAPHY

History of Cryptography

The history of cryptography can be traced back to the ancient Egyptians around 2000 BC, when they used complex and obscure hieroglyphic symbols to decorate tombs in order to secretly record the biographies, achievements, and contributions of the deceased.

For many centuries, one of the fields of study that attracted significant attention from scholars around the world was the effort to decipher the “hidden inscriptions” found within ancient Egyptian tombs. Through these studies, historians and researchers gained substantial insights into the history, customs, traditions, and daily life of the mysterious civilization of ancient Egypt.



Hieroglyphics – The Earliest Known Encryption Technique

ORIGINS OF CRYPTOGRAPHY

THE HISTORY OF CRYPTOGRAPHY:

The ancient Hebrews created a simple and efficient encryption algorithm known as the Atbash cipher, where the encryption and decryption key relies on an alphabetical substitution. Assuming a permutation table is used as the encryption :

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
↓																									
Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A

the plaintext "**JERUSALEM**" would be encrypted into the ciphertext "**QVIFHZOVN**".

Since this substitution cipher replaces only a single character (letter) with another single character, it is classified as a monoalphabetic substitution cipher. Alternatively, developers can design polyalphabetic substitution ciphers (or block-based substitutions). Instead of replacing individual characters, these algorithms substitute a sequence of plaintext characters with a corresponding sequence of ciphertext characters. This approach offers a significantly larger keyspace, thereby drastically reducing the vulnerability to cryptographic attacks.

ORIGINS OF CRYPTOGRAPHY

THE HISTORY OF CRYPTOGRAPHY

During Antiquity, the famous Roman emperor Julius Caesar designed a rudimentary encryption mechanism utilizing an algorithm known as a "**cyclic code**" (or more accurately, a **shift cipher**). Similar to the Hebrew Atbash cipher, this method relies on substitution; however, instead of an arbitrary mapping, it employs a **cyclic permutation**.



Caesar implemented this using two concentric disks, both inscribed with the Latin alphabet, where the inner disk corresponds to the plaintext and the outer disk represents the ciphertext. The encryption key is defined by rotating the outer disk a specific number of positions, shifting the alignment of the letters. For instance, if the key is +3 (a clockwise rotation of three positions), the plaintext letters A, B, C... X, Y, Z are mapped to the ciphertext letters D, E, F... A, B, C, respectively.

ORIGINS OF CRYPTOGRAPHY

THE HISTORY OF CRYPTOGRAPHY

The underlying principle of the Caesar shift cipher provided the conceptual foundation for developing the most renowned cryptographic device in history: the German **Enigma machine**, widely utilized during World War II. The Enigma machine integrated three mechanical **rotors**; upon entry, each character in the **plaintext** underwent a triple-layered substitution process governed by distinct, predefined algorithms, making **cryptanalysis** exceptionally formidable.



Subsequently, a group of Polish cryptologists successfully broke the Enigma cipher and provided the British with critical intelligence regarding German military operations. Historians estimate that this cryptanalytic breakthrough effectively shortened the duration of World War II by approximately two years. Following the conclusion of World War II, the operational secrets of the Enigma machine were declassified; today, an original Enigma machine is prominently exhibited at the Smithsonian Institution in Washington, D.C., USA.

ORIGINS OF CRYPTOGRAPHY

THE HISTORY OF CRYPTOGRAPHY

In 1922, William Frederic Friedman published his seminal work, "The Index of Coincidence and Its Applications in Cryptography". This publication established itself as one of the foundational standards for cryptographic nomenclature and classification. Consequently, Friedman is widely revered as a "titan" and "the father of modern American cryptanalysis" within the discipline of cryptography.



ORIGINS OF CRYPTOGRAPHY

MODERN CRYPTOGRAPHY

It is widely acknowledged that the era of modern cryptography began with Claude Shannon (April 30, 1916 – February 24, 2001), who is revered as the "father of mathematical cryptography." In 1949, he published his monumental paper, "Communication Theory of Secrecy Systems," in the Bell System Technical Journal. His profound contributions established the fundamental theoretical foundations for both cryptography and cryptanalysis.

Under this prevailing influence, cryptography became almost exclusively monopolized by government classified communications agencies, such as the NSA (National Security Agency), causing the discipline to vanish from the public eye. Consequently, very few academic works were openly published until the mid-1970s, a pivotal period when the cryptographic landscape underwent a dramatic transformation.



ORIGINS OF CRYPTOGRAPHY

CRYPTOGRAPHIC STANDARDS

The mid-1970s witnessed two major breakthroughs.

The first was the publication of the proposed Data Encryption Standard (DES) on March 17, 1975. This symmetric-key cipher was submitted by a research team at IBM, in response to an invitation by the National Bureau of Standards [NBS, now NIST]. The initiative aimed to develop secure electronic communication mechanisms for commercial enterprises, particularly banking institutions and large financial organizations. Consequently, DES became the first publicly accessible encryption standard.

In 2001, DES was officially superseded by the Advanced Encryption Standard (AES). This transition occurred after the 56-bit key length of DES succumbed to brute-force attacks powered by advancing computational strength, rendering the older standard structurally insecure for modern cryptographic needs.

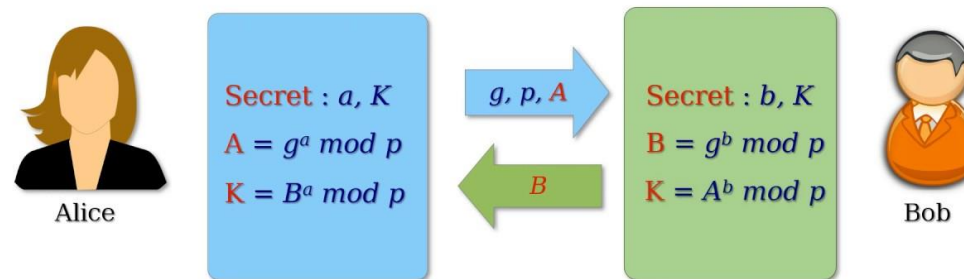


ORIGINS OF CRYPTOGRAPHY

CRYPTOGRAPHIC STANDARDS

The second advancement was even more monumental. In 1976, this breakthrough fundamentally altered the operational paradigm of cryptographic systems. It was catalyzed by the publication of the seminal paper, "New Directions in Cryptography," authored by Whitfield Diffie and Martin Hellman. The paper introduced an entirely unprecedented methodology for secure key distribution, known as the Diffie-Hellman key exchange. This publication served as a catalyst, stimulating the near-instantaneous evolution of a completely new class of encryption algorithms: asymmetric cryptography [or public-key cryptography].

Diffie - Hellman Key Exchange Protocol



THE CONCEPT OF MODERN CRYPTOGRAPHY

Principles of Modern Cryptography:

Modern cryptography is the process of encrypting digital data represented as sequences of binary digits [bits], unlike classical cryptography which operates directly on alphabetic characters. Modern cryptographic systems are designed to process these input binary streams [sequences] and transform them into alternative binary sequences.

Modern cryptography relies on publicly known, widely scrutinized mathematical algorithms to encrypt information. The security of these systems depends on the computational complexity [computational hardness] of the underlying mathematical problems, making it practically infeasible for an adversary to retrieve the original plaintext, even when the encryption algorithm itself is completely exposed.

Modern cryptographic algorithms are far too complex for manual human execution. Contemporary algorithms are executed exclusively by computers or specialized hardware devices, and in the vast majority of practical use cases, they are implemented via computer software.

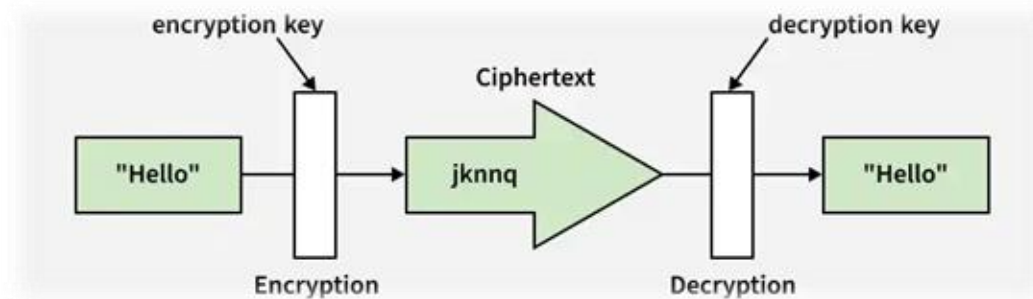


THE CONCEPT OF MODERN CRYPTOGRAPHY

Principles of Modern Cryptography:

Simply put, encryption is a method of protecting information by transforming data from an intelligible, readable format [plaintext] into an unintelligible, unreadable format [ciphertext] that can only be deciphered by authorized parties possessing the corresponding decryption key or password.

This practice significantly enhances data protection and ensures secure data transmission. In essence, data encryption does not prevent data from being intercepted or stolen; rather, it prevents unauthorized entities from understanding the content of the compromised files, as the information has been transformed into alternative character sets or structured formats. Encrypted data is formally referred to as ciphertext, whereas original, unencrypted data is termed plaintext.



THE CONCEPT OF MODERN CRYPTOGRAPHY

Modern cryptography is the cornerstone of computer security and communications. Its foundation relies on various mathematical concepts, including Number Theory, Computational Complexity Theory, Abstract Algebra, and Probability Theory.

There are three primary characteristics that distinguish modern cryptography from classical approaches:

Criteria	Classical Cryptography	Modern Cryptography
Data Manipulation	Operates directly on traditional characters, specifically letters and digits within natural language alphabets.	Operates on binary streams [sequences of binary digits / bits]. Modern cryptographic systems process these input binary streams and transform them into alternative binary sequences.
Security Principle	Primarily relies on " Security through obscurity. " The techniques, rules, or algorithmic designs used for encryption are kept secret, known only to the communicating parties.	Relies on publicly known, widely scrutinized mathematical algorithms to encrypt information. Security is achieved solely through the secrecy of the cryptographic key , which serves as the seed for the algorithm. The computational complexity [computational hardness] of the underlying mathematical problems makes it practically infeasible for an adversary to retrieve the original plaintext, even when the encryption algorithm itself is completely exposed. This strictly adheres to Kerckhoffs's Principle .
Key Ownership & Execution	Key Ownership & Execution Requires the entire cryptographic system (including the algorithm and operational mechanics) to remain a secret. The algorithms are simple enough for manual human execution using paper and pen.	Requires only the involved entities to possess and safeguard the secret key (or public/private key pairs). The algorithms are far too complex for manual human execution and must be executed exclusively by computers or specialized hardware devices, typically implemented via computer software.

THE CONCEPT OF MODERN CRYPTOGRAPHY

Cryptology, the study of secure communication systems, is fundamentally divided into two complementary branches:

- **Cryptography** – The branch of designing and constructing secure systems.
- **Cryptanalysis** – The branch of analyzing, evaluating, and breaking secure systems.



THE CONCEPT OF MODERN CRYPTOGRAPHY



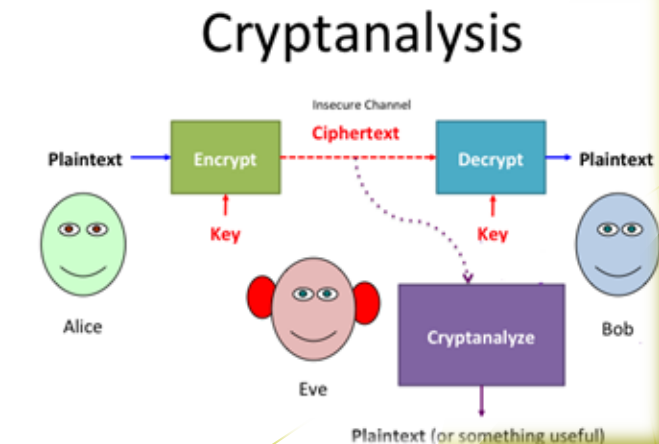
Cryptography:

- Cryptography is the art and science of engineering cryptosystems capable of providing robust information security services.
- In the digital era, cryptography addresses the safeguarding of digital data through the design of mechanisms anchored by complex mathematical algorithms. These cryptographic algorithms deliver core information security objectives, specifically: Confidentiality, Integrity, Authentication, and Non-repudiation. Cryptography can be conceptualized as a comprehensive toolkit containing diverse technical primitives utilized across various security applications.

Cryptanalysis:

- Cryptanalysis is the art and science of deciphering ciphertext and breaking cryptosystems without authorized access to the cryptographic keys.
- Cryptanalysis involves the rigorous study of cryptographic mechanisms with the intent of identifying mathematical weaknesses, logical flaws, or protocol vulnerabilities. Furthermore, it plays a vital role in the design lifecycle of new cryptographic algorithms, serving as an essential peer-review process to test their cryptographic strength, validate their security bounds, and prove their resilience against known adversarial attack models before standardization.

Cryptography focuses on the design and defense of secure systems, whereas Cryptanalysis investigates the examination, penetration, and validation of those very systems.



THE CONCEPT OF MODERN CRYPTOGRAPHY

CRYPTOGRAPHIC SECURITY SERVICES:

The primary objective of utilizing cryptography is to provide the following four fundamental information security services:

- **Confidentiality**
- **Data Integrity**
- **Authentication**
- **Non-repudiation**

CRYPTOGRAPHIC SECURITY SERVICES

Confidentiality:

Confidentiality is the fundamental security service provided by cryptography. It is a security service designed to keep information concealed from unauthorized entities. It is sometimes referred to as privacy or secrecy.

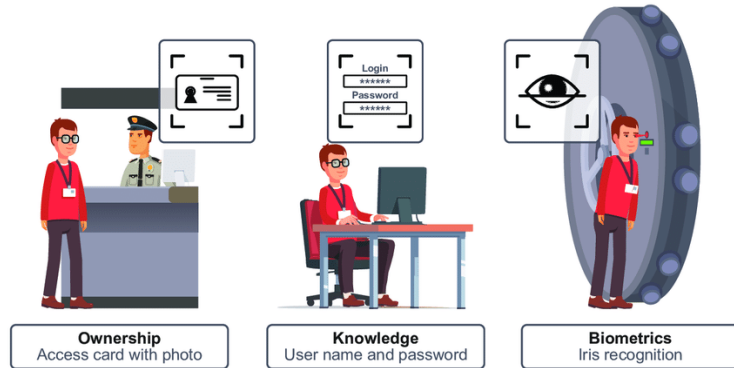


Data Integrity:

This security service is concerned with detecting any unauthorized modifications to data. Data can be altered by an unauthorized entity either intentionally or accidentally. The integrity service confirms whether the data has remained unaltered since it was last created, transmitted, or stored by an authorized entity.



CRYPTOGRAPHIC SECURITY SERVICES



Authentication:

Authentication provides the identification of the originator. It assures the receiver that the received data was genuinely sent by the identified and verified sender. The authentication service comprises two main variants:

- **Message authentication:** Identifies the origin of a message regardless of the routers or transmission systems that relayed it.
- **Entity authentication:** Ensures that the data is received from a specific entity in real-time, such as establishing a verified connection with a specific website.

Non-repudiation:

This security service ensures that an entity cannot deny ownership of a prior commitment or action. It guarantees that the original creator of the data cannot deny creating or transmitting that data to the recipient or a third party. Non-repudiation is a highly desirable property in scenarios where disputes over data exchange are likely to occur.



CRYPTOGRAPHIC SECURITY SERVICES

CRYPTOGRAPHIC PRIMITIVES:

Cryptographic primitives are the fundamental tools and techniques in cryptology. They can be selectively utilized to provide a desired set of information security services:

- **Encryption**
- **Hash Functions**
- **Message Authentication Codes (MACs)**
- **Digital Signatures**

Cryptographic Primitive	Symmetric/Asymmetric Encryption	Hash Functions	Message Authentication Codes (MAC)	Digital Signatures
Confidentiality	Yes	No	No	No
Data Integrity	No	Yes	Yes	Yes
Authentication	No	No	Yes	Yes
Non-repudiation	No	No	No	Yes

Cryptographic primitives are intricately interrelated, and they are frequently combined to achieve a comprehensive, desired set of security services within a robust cryptosystem.

Fundamental Primitives of Modern Cryptography

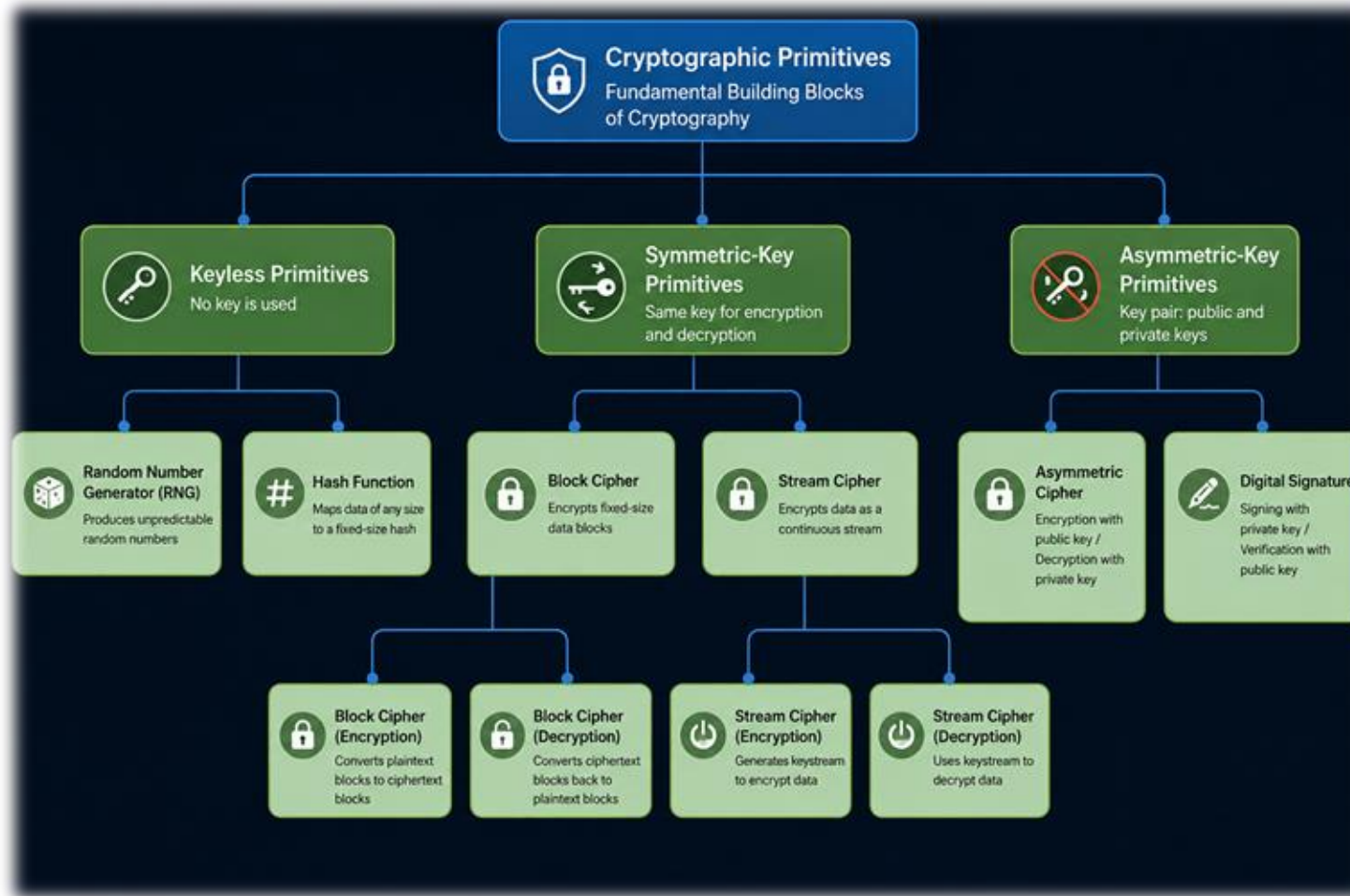


Diagram of the basic elements of cryptography

Challenges and Limitations of Modern Cryptographic Systems

In addition to the four fundamental objectives of information security, several other issues significantly affect the effective and practical use of secure information systems.

Limitations of Modern Cryptography:

- Information that is strongly encrypted, authenticated, and digitally signed may become difficult to access even for legitimate users at critical moments. Furthermore, computer networks or systems may still be compromised or disrupted by attackers.
- High availability, one of the core principles of information security, cannot be guaranteed solely through cryptographic mechanisms. Additional protection methods are required to defend against threats such as Denial-of-Service (DoS) attacks, hardware failures, or complete system outages.
- Another essential security requirement, selective access control, also cannot be achieved through cryptography alone. Administrative policies, operational procedures, and access management mechanisms must be implemented in conjunction with cryptographic techniques.
- Cryptography does not protect against vulnerabilities and threats arising from poor system architecture, insecure protocols, software bugs, or improper operational procedures. Such weaknesses must be addressed through secure system design and the establishment of a robust defense infrastructure.
- Computational and Financial Costs of Cryptography
 - The integration of cryptographic techniques into information-processing systems introduces computational overhead, resulting in additional latency and reduced system performance.
 - The deployment of public-key cryptography requires the establishment and maintenance of a Public Key Infrastructure (PKI), which can involve substantial financial and administrative costs.

FUTURE OF MODERN CRYPTOGRAPHIC SYSTEMS

The future of modern cryptographic systems is significantly influenced by the advancement of **Quantum Computing**, which is considered one of the greatest threats to traditional cryptographic algorithms.

In classical computers, data is represented in binary form using **bits**, where each bit can store only one of two possible values: 0 or 1. In contrast, quantum computers utilize **qubits (quantum bits)**, which can exist simultaneously in multiple states through the phenomenon of **quantum superposition**. Furthermore, qubits may interact through **quantum entanglement**, enabling massive parallel computational capabilities.

Due to these properties, quantum computers can solve certain computational problems exponentially faster than classical transistor-based computers. This poses a serious threat to public-key cryptographic systems such as RSA, Diffie–Hellman, and Elliptic Curve Cryptography (ECC), all of which rely on mathematical problems that are computationally infeasible for classical computers.

For example, the security of RSA is based on the difficulty of factoring large integers into prime numbers. Historically, RSA-640 (a 193-digit RSA number) required a large cluster of classical computers operating continuously for several months to be factored successfully. However, a sufficiently powerful quantum computer running **Shor's algorithm** could theoretically perform the same task within seconds or minutes.

The rapid progress of quantum computing may eventually weaken or completely break many currently deployed cryptographic systems. Consequently, the information security research community is actively developing **Post-Quantum Cryptography (PQC)**, which consists of cryptographic algorithms designed to resist attacks from quantum computers while remaining compatible with existing computing infrastructure.

POST-QUANTUM CRYPTOGRAPHY

The advancement of quantum computing introduces unprecedented computational capabilities, posing significant challenges to the foundations of modern cryptography.

The security of most existing cryptographic systems — including data encryption, authentication, digital signatures, e-commerce, online auctions, and electronic voting — relies on the assumption that certain mathematical problems are computationally infeasible for classical computers to solve within a practical timeframe. However, the emergence of quantum computers threatens many of these security assumptions.

For instance:

Shor's algorithm can efficiently solve integer factorization and discrete logarithm problems.

As a result, widely deployed public-key cryptosystems such as RSA, Diffie–Hellman, and Elliptic Curve Cryptography (ECC) may become vulnerable.

Consequently, the information security community is actively developing new cryptographic approaches resistant to quantum attacks. Two major directions currently receiving significant attention are:

1. Quantum-Safe Cryptography / Post-Quantum Cryptography (PQC)
2. Quantum Key Distribution (QKD)



POST-QUANTUM CRYPTOGRAPHY

1. Quantum-Safe Cryptography / Post-Quantum Cryptography (PQC)

Post-Quantum Cryptography (PQC) refers to cryptographic algorithms designed to remain secure against both classical and quantum computers.

PQC algorithms are based on mathematical problems for which no efficient quantum algorithms are currently known, including:

- Lattice-based cryptography,
- Code-based cryptography,
- Hash-based cryptography,
- Multivariate cryptography.

A major advantage of PQC is that it can be implemented on existing classical computing infrastructure without requiring specialized quantum hardware.

Currently, National Institute of Standards and Technology (NIST) is leading the standardization process for post-quantum cryptographic algorithms intended to replace traditional public-key cryptosystems in the future.

2. Quantum Key Distribution (QKD)

Quantum Key Distribution is a method that utilizes the principles of quantum mechanics to establish a shared secret key between two trusted parties.

Unlike traditional cryptographic systems, QKD does not rely on computational hardness assumptions. Instead, it is based on quantum mechanical properties such as:

- The Heisenberg Uncertainty Principle,
- Quantum superposition,
- And the No-Cloning Theorem.

The primary advantage of QKD is that any eavesdropping attempt will disturb the quantum states being transmitted, thereby revealing the presence of an attacker immediately.

However, QKD requires specialized communication infrastructure, including:

- High-quality optical fiber channels,
- Photon detection devices,
- Or quantum communication satellites,

which currently makes large-scale deployment expensive and technically challenging.

QUANTUM CRYPTOGRAPHY

Quantum Cryptography is a field that applies the principles of quantum mechanics to secure information and transmit data with an extremely high level of security. Unlike classical cryptography, which relies on the computational difficulty of mathematical problems, quantum cryptography is based on the laws of quantum physics to detect and prevent eavesdropping attempts.

The most prominent application of quantum cryptography today is **Quantum Key Distribution (QKD)**, which enables two parties to establish a shared secret key while ensuring that any interception attempt can be detected.

Although the concept appears straightforward, the true complexity of quantum cryptography lies in the underlying principles of quantum mechanics, including:

Fundamental Principles of Quantum Cryptography

- **Quantum Superposition:** A quantum particle can exist simultaneously in multiple states until it is observed or measured.
- **Photon Polarization States:** In QKD systems, photons are randomly generated in different polarization states to represent quantum information.
- **Heisenberg Uncertainty Principle:** Measuring a quantum state inevitably disturbs or alters that state. Therefore, if an attacker attempts to intercept and measure transmitted quantum data, the intrusion introduces detectable disturbances.
- **No-Cloning Theorem:** It is impossible to create an identical copy of an unknown quantum state. This property prevents attackers from perfectly duplicating secret quantum keys during transmission.
- **Quantum Entanglement:** Two or more quantum particles may become entangled such that the state of one particle is directly correlated with the state of another, even across large distances. This phenomenon is utilized in advanced quantum communication protocols.
- Together, these principles form the foundation of quantum cryptography, where security is guaranteed not only by algorithms but also by the fundamental laws of physics.



Example of How Quantum Cryptography Works

Alice and Bob want to exchange confidential information without allowing any third party to intercept their communication. To achieve this, they use **Quantum Key Distribution (QKD)** over an optical communication channel such as fiber optics or free-space optical links.

In this process, Alice transmits a sequence of photons encoded with randomly chosen quantum polarization states. Each photon represents a quantum bit of information and can only be measured correctly using the appropriate quantum basis.

Because photons exist in quantum states, any measurement attempt inevitably disturbs those states. If an eavesdropper (commonly called Eve) attempts to intercept the transmission:

- Eve must measure the quantum state of each photon,
- The measurement process alters the original quantum states according to the principles of quantum mechanics,
- When Bob receives the photons and compares a subset of the transmission with Alice, they can detect an abnormal error rate in the generated quantum key.

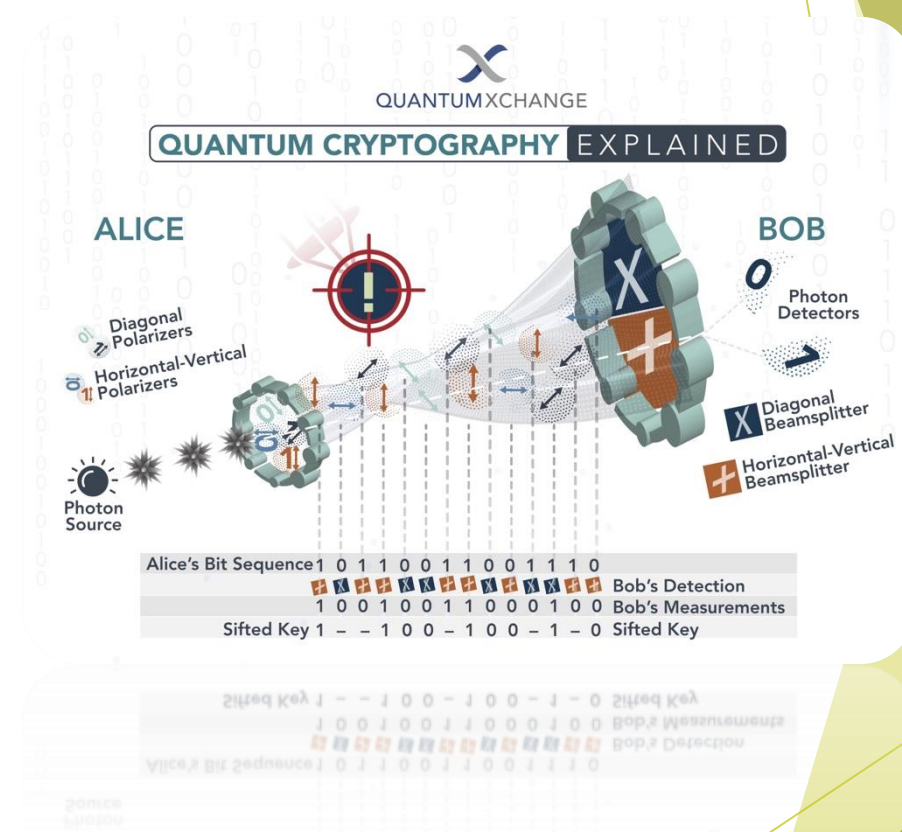
If the error rate exceeds an acceptable threshold, Alice and Bob conclude that the communication channel has been compromised. In that case:

- The quantum key is discarded,
- A new key generation process is initiated,
- And only a verified secure key is used for actual encryption and decryption.

The key advantage of quantum cryptography is that:

- Security does not rely solely on computational complexity,
- But on the fundamental laws of physics, where any eavesdropping attempt introduces detectable physical disturbances.

As a result, QKD enables highly secure key exchange, even against future attacks from powerful quantum computers.

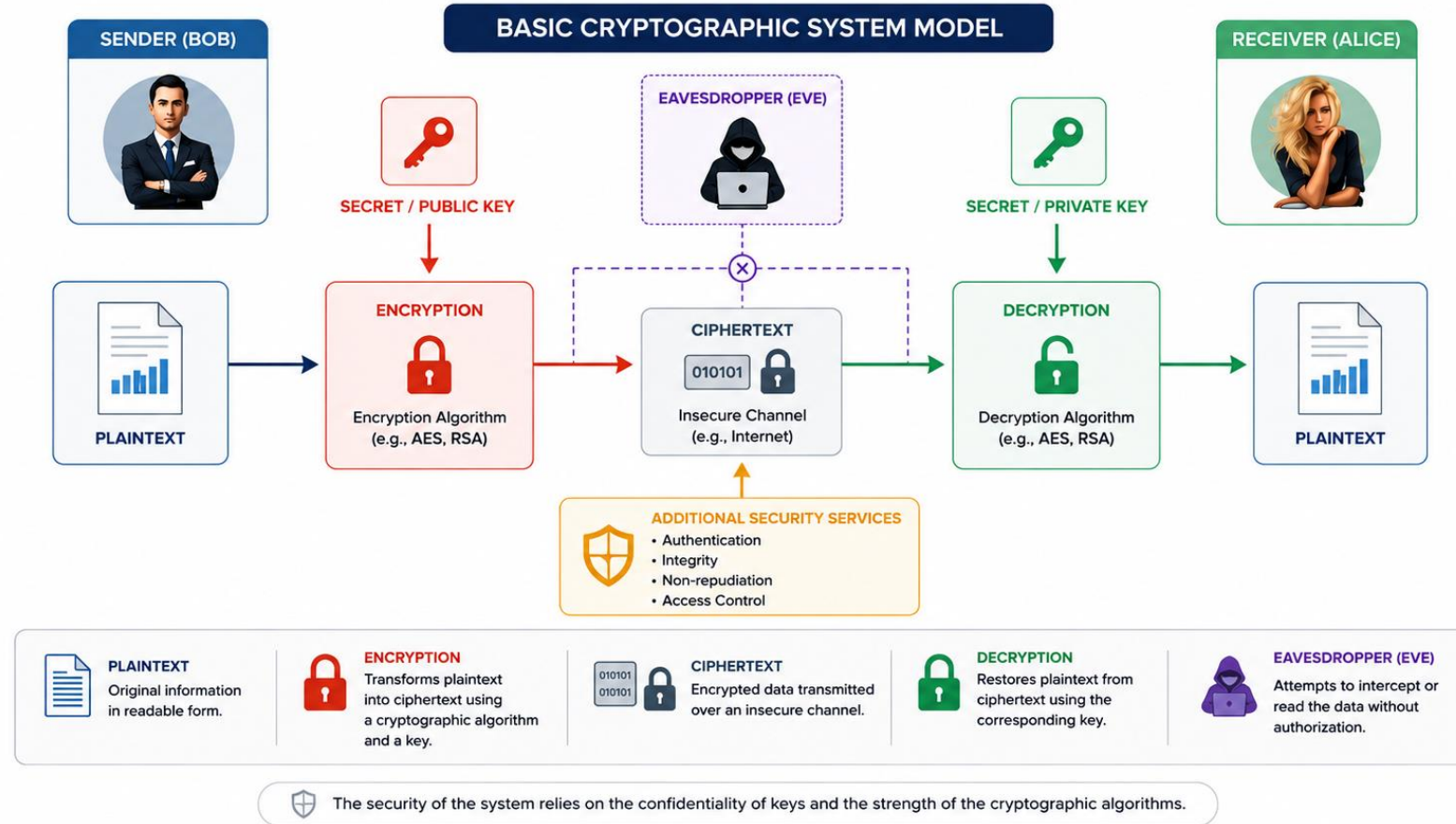


CLASSICAL CRYPTOGRAPHY VS QUANTUM CRYPTOGRAPHY

Classical Cryptography	Quantum Cryptography
Based on digital computing logic and binary systems.	Based on the principles of quantum mechanics.
Data is represented and transmitted using binary bits (0 and 1).	Data is transmitted using quantum particles, typically photons or qubits.
Security primarily relies on the computational hardness of mathematical problems.	Security relies on the laws of quantum physics.
Uses algorithms such as AES, RSA, ECC, and SHA.	Uses protocols such as Quantum Key Distribution (QKD).
Can operate on existing Internet and network infrastructure.	Typically requires specialized communication infrastructure such as quantum fiber-optic channels or optical links.
Generally does not have strict physical distance limitations.	Transmission distance is limited by photon loss and may require quantum repeaters.
Potentially vulnerable to future quantum computer attacks.	Can detect eavesdropping attempts through quantum state disturbances.
Eavesdropping may remain undetected if cryptographic keys are compromised.	Any measurement or interception attempt introduces detectable disturbances.
Lower deployment cost and widely standardized.	Higher deployment cost due to specialized hardware and infrastructure requirements.
Widely used in modern information systems today.	Currently applied mainly in high-security research and specialized communication systems.

CRYPTOGRAPHIC SYSTEM

A cryptographic system is the implementation of cryptographic techniques and supporting infrastructure to provide information security services.



TỔNG QUAN VỀ MẬT MÃ

HỆ THỐNG MẬT MÃ

Các thành phần của hệ thống mật mã:

- **Bản rõ (plaintext):** dữ liệu cần được bảo vệ trong quá trình truyền.
- **Bản mã (ciphertext):** phiên bản được xáo trộn của bản rõ, được tạo ra bởi thuật toán mã hóa bằng cách sử dụng một khóa mã hóa cụ thể. Bản mã không được bảo vệ. Nó truyền trên kênh công cộng (public channel). Nó có thể bị chặn hoặc xâm phạm bởi bất kỳ ai có quyền truy cập vào kênh liên lạc.
- **Thuật toán mã hóa(Encryption algorithm):** đó là một quá trình toán học tạo ra một bản mã cho bất kỳ khóa mã hóa và bản rõ nào. Nó là thuật toán mã hóa lấy văn bản gốc và khóa mã hóa làm đầu vào để tạo ra một bản mã.
- **Thuật toán giải mã(Decryption algorithm):** Đây là một quá trình toán học, tạo ra một bản rõ duy nhất từ bất kỳ bản mã và khóa giải mã đã cho. Nó là một thuật toán mã hóa lấy bản mã và khóa giải mã làm đầu vào để xuất ra bản rõ. Thuật toán giải mã về cơ bản là đảo ngược của thuật toán mã hóa.
- **Khóa mã hóa(Encryption key):** Đó là một giá trị mà người gửi đã biết. Người gửi nhập khóa mã hóa vào thuật toán mã hóa cùng với bản rõ để tính toán ra bản mã.
- **Khóa giải mã(Decryption key):** là một giá trị được biết bởi người nhận. Khóa giải mã có liên quan đến khóa mã hóa, nhưng không phải lúc nào cũng giống với khóa mã hóa. Người nhận nhập khóa giải mã vào thuật toán giải mã cùng với bản mã để tính toán ra bản rõ.

Đối với một hệ thống mật mã nhất định, một tập hợp tất cả các khóa giải mã có thể được gọi là **không gian khóa (key space)**.

Kẻ tấn công(chặn – interceptor) là một thực thể trái phép cố gắng xác định bản rõ (plaintext). Kẻ tấn công có thể xem bản mã và có thể biết thuật toán giải mã. Tuy nhiên interceptor không bao giờ biết được khóa giải mã.

SYMMETRIC ENCRYPTION

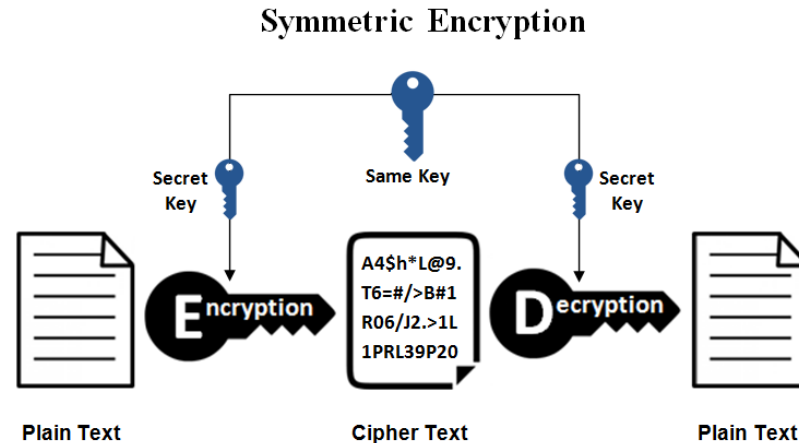
Symmetric-Key Encryption is a cryptographic process in which the same secret key is used for both encryption and decryption of information. This means that both the sender and the receiver must securely share and protect the same secret key before communication takes place.

In a symmetric cryptosystem:

- Plaintext is transformed into ciphertext using an encryption algorithm and a secret key.
- The receiver uses the same key to recover the original plaintext.

Well-Known Symmetric Encryption Algorithms

- DES (Data Encryption Standard)
- Triple DES (3DES)
- AES – Rijndael (Advanced Encryption Standard)
- IDEA (International Data Encryption Algorithm)
- Blowfish
- Twofish
- RC4 (stream cipher – no longer recommended)
- ChaCha20



Before the 1970s, nearly all cryptographic systems relied exclusively on symmetric encryption because public-key cryptography had not yet been invented.

Even today, symmetric encryption remains a fundamental component of modern information security. In practice, most Internet data traffic is still encrypted using symmetric algorithms such as AES or ChaCha20 due to their high performance and efficiency.

Due to its computational efficiency and high performance, symmetric encryption is expected to remain a core technology in future secure communication systems, including the post-quantum era.

SYMMETRIC ENCRYPTION

Characteristics of Symmetric Encryption

Advantages

- Very fast encryption and decryption speed.
- Highly efficient for processing large volumes of data.
- Requires less computational power compared to asymmetric cryptography.
- Suitable for:
 - VPNs,
 - SSL/TLS,
 - disk encryption,
 - database protection,
 - real-time secure communication.

Disadvantages

- The primary challenge is the secure distribution of the secret key between communicating parties.
- If the secret key is compromised, all encrypted data can be decrypted.
- Key management becomes difficult in large-scale systems with many users.

ASYMMETRIC ENCRYPTION

Asymmetric-Key Encryption is a cryptographic process in which two different but mathematically related keys are used:

- One key is used for encryption,
- And the other key is used for decryption.

These two keys are known as:

- **Public Key:** can be openly distributed.
- **Private Key:** must be kept secret by the owner.

In an asymmetric cryptosystem:

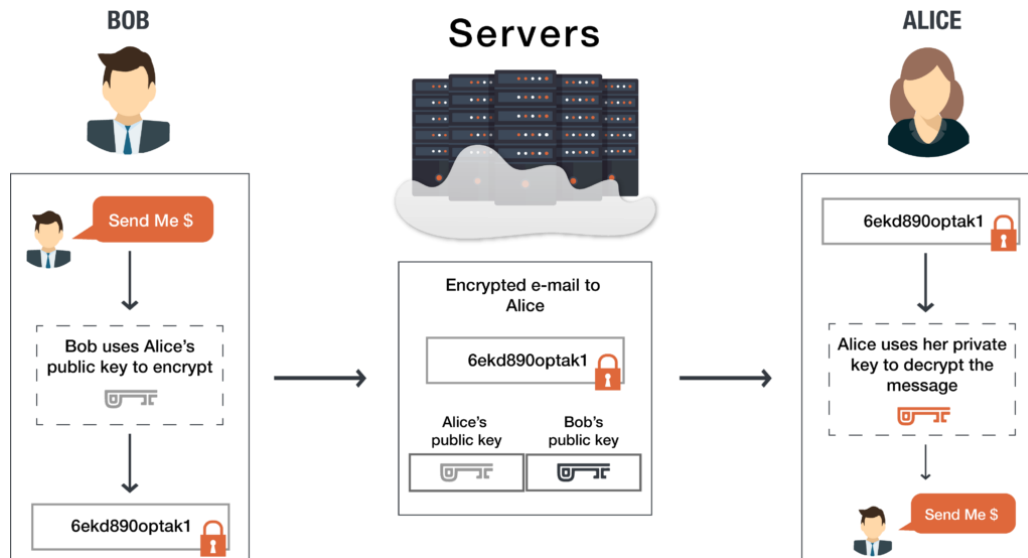
- Data is encrypted using the recipient's public key,
- And only the corresponding private key can decrypt the ciphertext.

Although the keys are different, they are mathematically linked in such a way that:

- Encryption and decryption remain possible,
- While deriving the private key from the public key is computationally infeasible.

Well-Known Asymmetric Cryptographic Algorithms

- RSA (Rivest–Shamir–Adleman)
- Diffie–Hellman Key Exchange
- ElGamal
- Elliptic Curve Cryptography (ECC)
- DSA (Digital Signature Algorithm)
- ECDSA (Elliptic Curve Digital Signature Algorithm)



ASYMMETRIC ENCRYPTION

Characteristics of Asymmetric Encryption

Advantages

Eliminates the need to share a secret key in advance.

Supports:

- secure encryption,
- authentication,
- digital signatures,
- and secure key exchange.

Provides better scalability for large networks and Internet-based systems.

Disadvantages

Slower than symmetric encryption.

Requires greater computational resources.

Not efficient for directly encrypting large amounts of data.

Real-World Applications

Asymmetric cryptography is widely used in:

- HTTPS/TLS,
- Digital signatures,
- Digital certificates,
- Public Key Infrastructure (PKI),
- Secure email systems (PGP, S/MIME),
- Blockchain and cryptocurrencies,
- VPNs and SSH.

Public Key Infrastructure (PKI)

PKI is a framework responsible for:

- key generation,
- key distribution,
- public key authentication,
- certificate revocation,
- and digital identity management.

In this model, a trusted third party known as a:

- **Certificate Authority (CA)**

is responsible for verifying user identities and issuing digital certificates.

Digital Certificates

Public keys are commonly embedded within digital certificates that are digitally signed by a trusted CA.

A digital certificate typically contains:

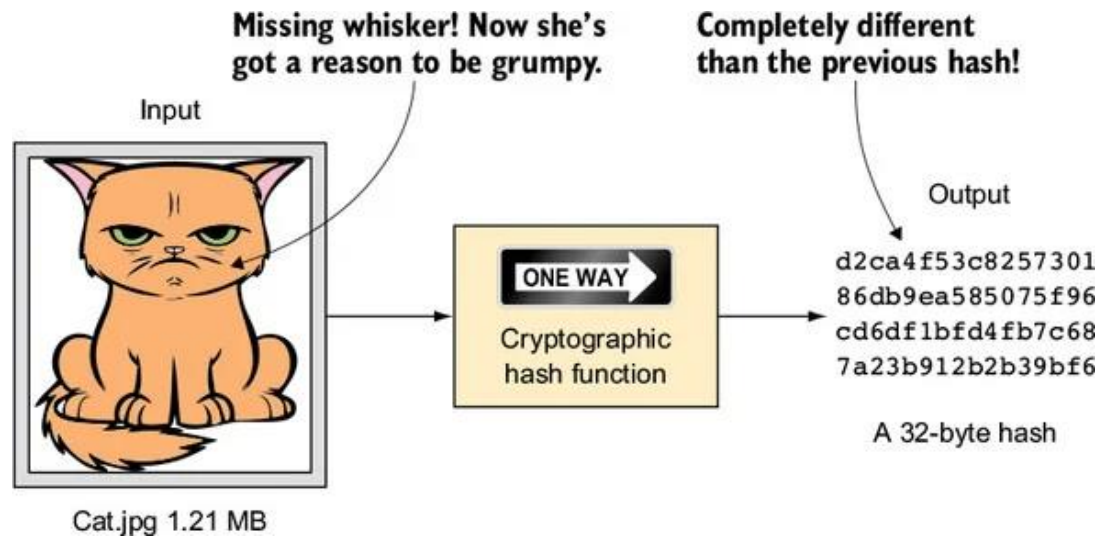
- Owner identification information,
- Public key,
- Validity period,
- Cryptographic algorithms used,
- Digital signature of the CA.

When users receive a public key:

- They verify the CA's digital signature,
- If the signature is valid, the public key is considered trustworthy.

HASH FUNCTION

Hash Function: A hash function is a mathematical function that transforms an input value into another compressed numerical value called a hash value or digest. The input to a hash function may have arbitrary length, while the output always has a fixed length. A hash function is also referred to as a one-way function because it supports only the hashing process and not decryption, as no cryptographic key is involved.



CRYPTOGRAPHIC HASH FUNCTIONS

Typical Features of Hash Functions

Fixed-Length Output (Hash Value / Message Digest)

A hash function transforms input data of arbitrary length into a fixed-length output known as a **hash value** or **message digest**.

This process is called **hashing**.

Since the hash value is typically much smaller than the original input data, hash functions are sometimes considered mathematical compression functions. However, unlike traditional compression algorithms, the original data cannot be reconstructed from the hash value.

A hash function that produces an output of n bits is called an **n -bit hash function**. Common modern hash functions typically generate outputs of:

- 160 bits (SHA-1),
- 256 bits (SHA-256),
- 384 bits,
- or 512 bits (SHA-512).

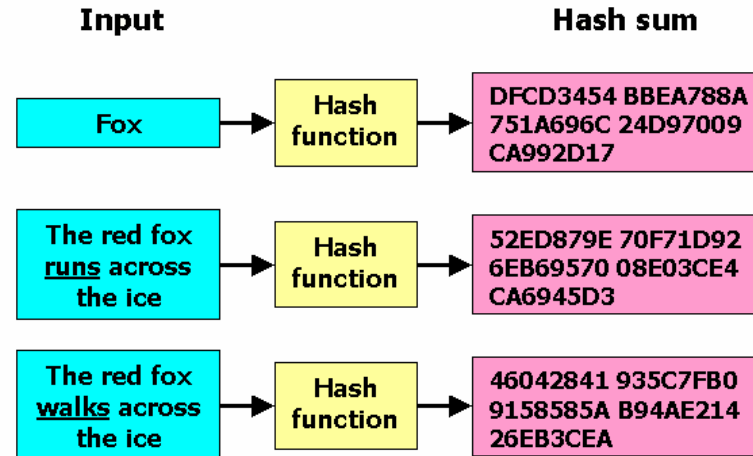
Computational Efficiency

For a hash function $h(x)$:

- Computing the hash value is generally very fast and computationally efficient.
- Hash functions are usually significantly faster than symmetric encryption algorithms.

This efficiency makes hash functions suitable for:

- data integrity verification,
- authentication,
- digital signatures,
- blockchain systems,
- and password storage.



CRYPTOGRAPHIC HASH FUNCTIONS

1. Preimage Resistance

Given a hash value $h(x)$, it should be computationally infeasible to recover the original input x .

This property represents the one-way nature of cryptographic hash functions.

2. Second Preimage Resistance

Given an input x_1 , it should be computationally infeasible to find another input $x_1 \neq x_2$ such that:

$$h(x_1) = h(x_2)$$

This property helps preserve the uniqueness of input data.

3. Collision Resistance

It should be computationally infeasible to find two different inputs that produce the same hash value.

In other words, finding:

$$x_1 \neq x_2 \text{ such that } h(x_1) = h(x_2)$$

must be practically impossible.

This property is critically important in:

- digital signatures,
- digital certificates,
- blockchain technologies,
- and data integrity verification.

EVALUATING THE SECURITY OF A CRYPTOSYSTEM

Evaluating the security of a cryptosystem is the process of analyzing whether the system can adequately protect information against various types of attacks. This is one of the most critical issues in modern information security and cryptography.

A cryptographic system is considered **insecure** if there exists a feasible attack method that allows an attacker to:

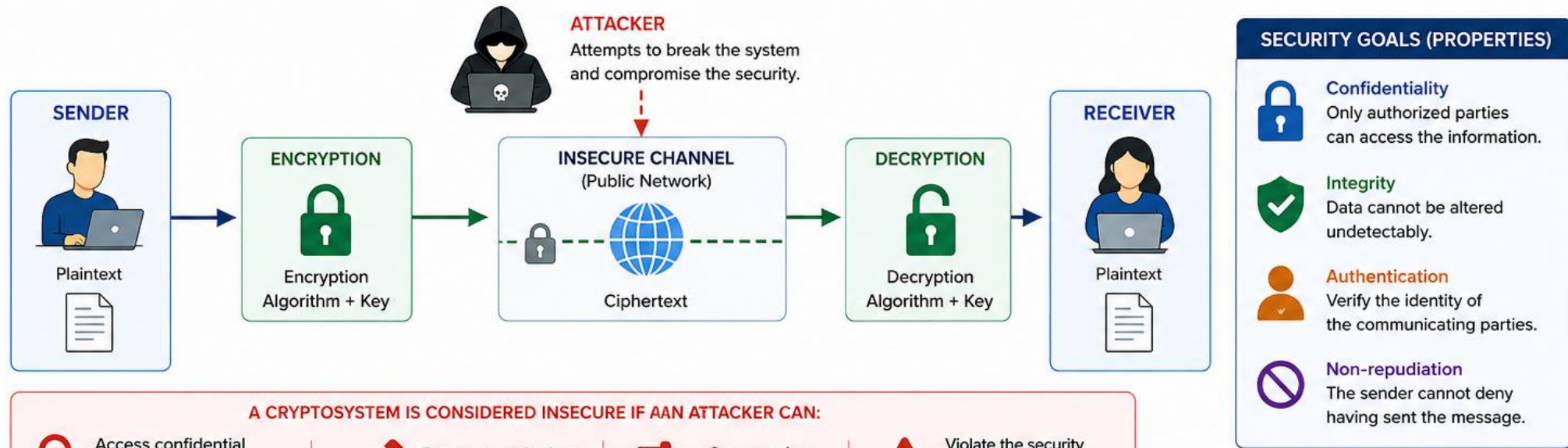
- access confidential information,
- forge or manipulate data,
- compromise authentication mechanisms,
- or violate the security properties that the system is intended to guarantee.

However, proving that a cryptosystem is “absolutely secure” is generally impossible in practice. Instead, cryptographic systems are typically evaluated based on:

- the computational difficulty of breaking the system,
- the computational resources and time required to perform an attack,
- and the system’s resistance against known attack models and cryptanalytic techniques.

EVALUATING THE SECURITY OF A CRYPTOSYSTEM

The process of analyzing whether a cryptosystem can adequately protect information against various types of attacks.



A CRYPTOSYSTEM IS CONSIDERED INSECURE IF AN ATTACKER CAN:



Access confidential information (e.g., recover plaintext or secret key)



Forge or manipulate data (violate integrity)

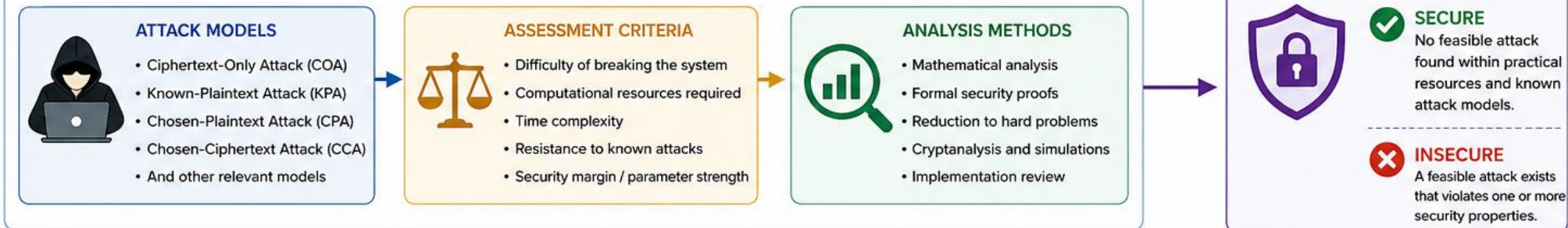


Compromise authentication mechanisms



Violate the security properties the system is intended to guarantee

HOW SECURITY IS EVALUATED



Absolute security is unattainable in practice. The goal is to make breaking the cryptosystem computationally infeasible with current technology and resources.

CIPHERTEXT-ONLY ATTACK - COA

A Ciphertext-Only Attack (COA) is the most fundamental attack model in cryptography, in which the attacker has access only to ciphertexts transmitted over a communication channel, without knowing:

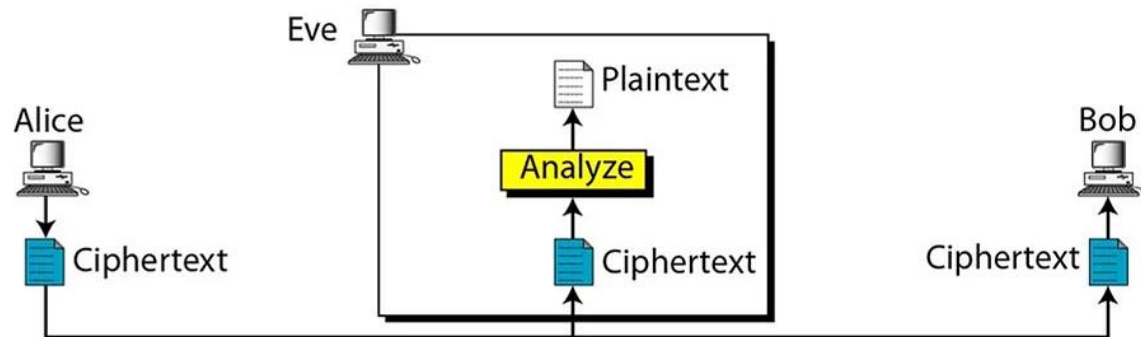
- the corresponding plaintexts,
- the secret key,
- or any internal system information.

In this model:

- Alice sends encrypted messages to Bob,
- The attacker, Eve, can only eavesdrop on the communication channel and collect ciphertexts Y.

Eve's objective may include:

- recovering the plaintext X,
- deriving the secret key Z,
- or completely breaking the cryptosystem.



This attack model represents the minimum security requirement that every modern cryptosystem must resist. If a cryptosystem cannot withstand a Ciphertext-Only Attack, it is considered insecure and unreliable in practice.

CIPHERTEXT-ONLY ATTACK - COA

Characteristics of Ciphertext-Only Attack

1. The attacker has very limited information

Eve knows only:

- the encryption algorithm (according to Kerckhoffs's Principle),
- and the intercepted ciphertexts.

Eve does not know:

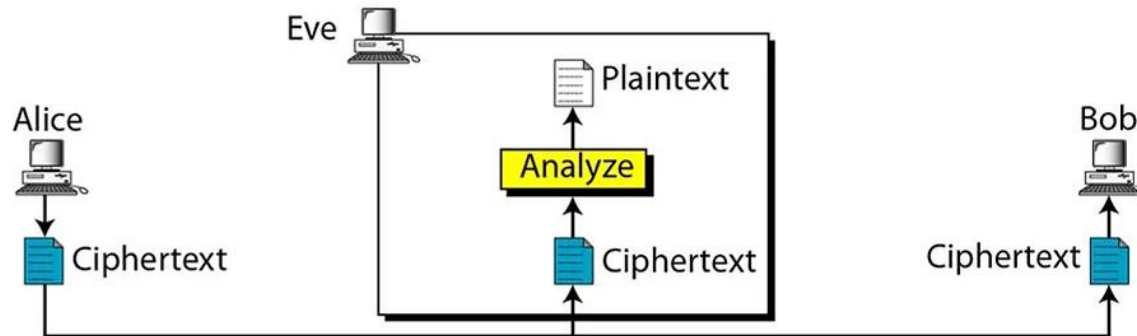
- the secret key,
- the plaintexts,
- or the internal implementation details.

2. Statistical analysis is commonly used

In classical cryptography, attackers often rely on:

- character frequency analysis,
- repeated patterns,
- language structures,
- or statistical properties of ciphertexts

to infer plaintexts or secret keys.



KNOWN-PLAINTEXT ATTACK – KPA

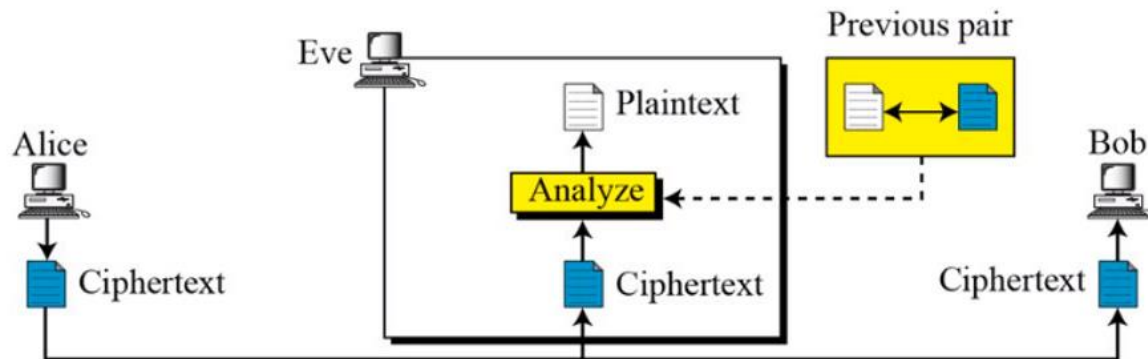
A Known-Plaintext Attack (KPA) is an attack model in which the attacker has access to one or more pairs of:

- plaintext X,
- and the corresponding ciphertext Y.

In other words, unlike a Ciphertext-Only Attack, the attacker not only intercepts ciphertexts but also knows some of the original plaintexts associated with them.

The attacker's objective remains:

- recovering additional secret plaintexts,
- deriving the secret key,
- or completely compromising the cryptosystem.



How Can the Attacker Obtain Plaintexts?

In practice, obtaining some plaintext information is quite realistic. This may occur due to:

- standardized document formats,
- predictable message headers,
- publicly known communication protocols,
- repeated data patterns,
- insider leaks,
- or compromised low-level employees within the system.

KNOWN-PLAINTEXT ATTACK – KPA

Why is KPA Stronger than Ciphertext-Only Attack?

In a Ciphertext-Only Attack:

Eve has access only to ciphertexts.

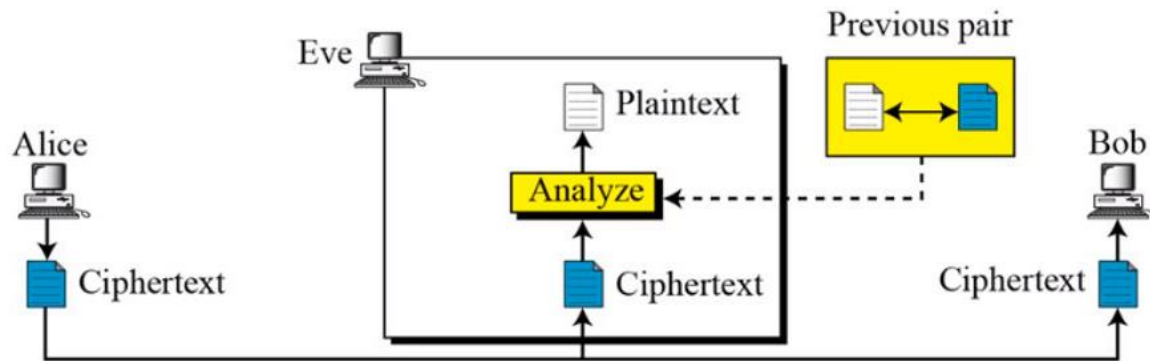
In a Known-Plaintext Attack:

Eve additionally knows plaintext-ciphertext pairs:

$$(X, Y) \text{ such that } \text{Enc}(K, X) = Y$$

This allows Eve to:

- test candidate keys,
- eliminate incorrect keys,
- and significantly reduce the key search space.



Exhaustive key search in KPA

A common attack technique is:

- Exhaustive Key Search (Brute Force Attack).

Eve proceeds by:

1. Trying candidate keys K ,
2. Encrypting the known plaintext X ,
3. Comparing the result with the known ciphertext Y .

If:

$$\text{Enc}(K, X) = Y$$

then K may be the correct key.

The more matching plaintext-ciphertext pairs available:

- the higher the confidence in the recovered key.

CHOSEN-PLAINTEXT ATTACK – CPA

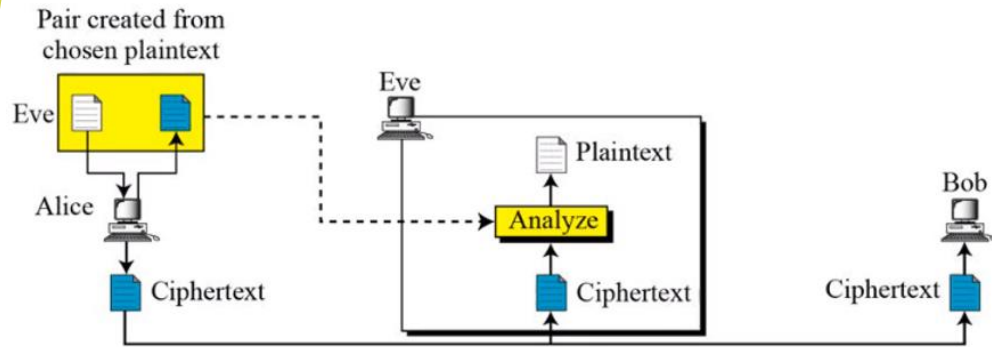
A Chosen-Plaintext Attack (CPA) is an attack model in which the attacker not only knows some plaintext-ciphertext pairs:

- plaintext X,
- and corresponding ciphertext Y,

but can also **choose arbitrary plaintexts** and obtain their corresponding ciphertexts from the encryption system.

In other words, Eve has access to an **“encryption oracle”**:

- Eve submits chosen plaintexts,
- and the system returns the corresponding ciphertexts.



Why Is CPA Realistic?

Although CPA may initially seem unrealistic, it is highly relevant in practice.

Examples include:

- insider threats,
- legitimate user accounts,
- public encryption services,
- automatic encryption systems,
- web applications,
- cloud encryption APIs,
- and network protocols.

In these environments:

- attackers may intentionally submit carefully crafted plaintexts to analyze system behavior.

CHOSEN-PLAINTEXT ATTACK – CPA

Why Is CPA Stronger Than KPA?

In a Known-Plaintext Attack:

- Eve only observes existing plaintexts.

In CPA:

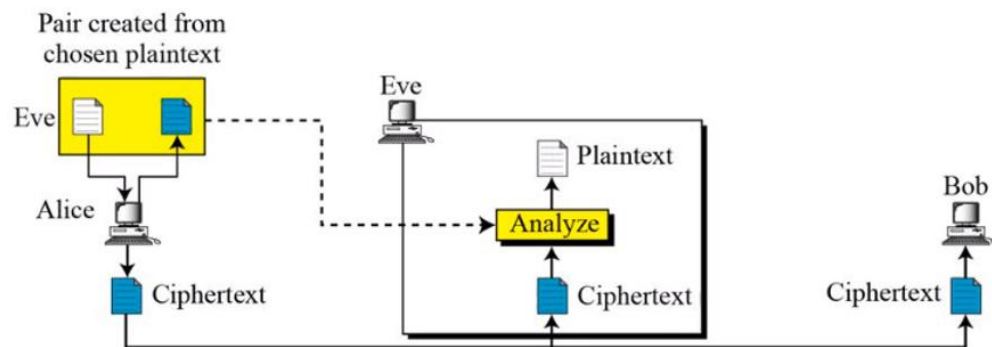
- Eve actively designs plaintexts to exploit cryptographic behavior.

For example:

- repeated patterns,
- all-zero blocks,
- structured inputs,
- or controlled bit modifications.

This allows Eve to:

- analyze relationships between plaintext and ciphertext,
- detect structural weaknesses,
- infer key-related information,
- or exploit implementation flaws.



Adaptive Chosen-Plaintext Attack

A stronger variant is: Adaptive CPA.

In this model: Eve selects new plaintexts based on previous encryption results.

This enables:

- dynamic cryptanalysis,
- optimized attack strategies,
- and more effective key recovery attempts.

Importance in Modern Cryptography

CPA is one of the most important security models in modern cryptography.

A cryptosystem is considered secure only if:

- ciphertexts reveal no useful information,
- even when attackers can choose arbitrary plaintexts.

This security notion is known as:

- **IND-CPA Security**
(Indistinguishability under Chosen-Plaintext Attack).

CHOSEN-CIPHERTEXT ATTACK – CCA

A Chosen-Ciphertext Attack (CCA) is one of the strongest attack models in modern cryptography.

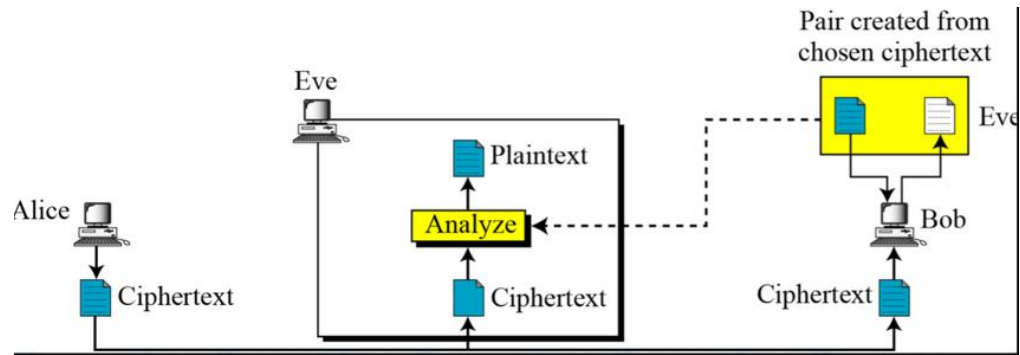
In this attack model:

- the attacker, Eve, is capable of selecting arbitrary ciphertexts,
- submitting them to a decryption system,
- and obtaining the corresponding plaintexts.

In other words, Eve has access to a: **decryption oracle**.

However:

- Eve is not allowed to directly request the decryption of the target ciphertext that she intends to break.
- Instead, the goal is to use the decryption results of other chosen ciphertexts to derive:
 - confidential plaintext information,
 - the secret key,
 - or other useful cryptographic information.



Why Is CCA Dangerous?

CCA is particularly dangerous because:

- the attacker can actively manipulate ciphertexts,
- observe the system's responses,
- and exploit weaknesses in cryptographic algorithms or protocol implementations.

This enables significantly deeper cryptanalysis compared to CPA, especially in systems that reveal different error responses during decryption.

CHOSEN-CIPHERTEXT ATTACK – CCA

Adaptive Chosen-Ciphertext Attack (CCA2)

A stronger variant is: Adaptive Chosen-Ciphertext Attack (CCA2).

In this model: the attacker may continue selecting new ciphertexts based on previous decryption results.

CCA2 is widely regarded as: one of the highest security standards in modern cryptography.

Importance in Modern Cryptography

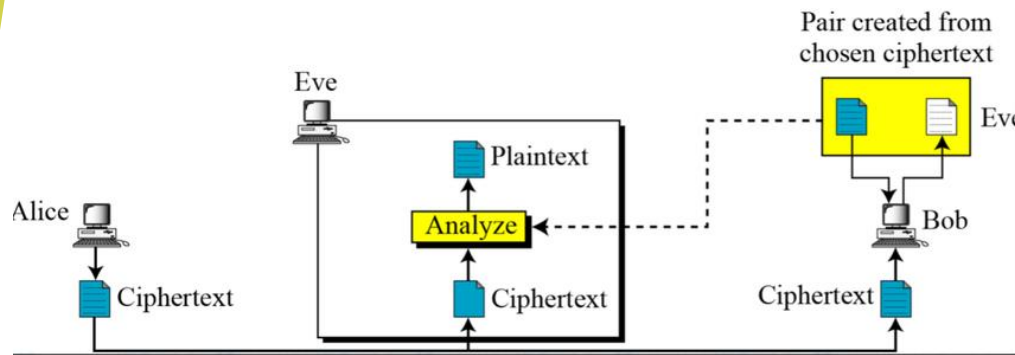
A strong modern cryptosystem should achieve:

IND-CCA Security

(Indistinguishability under Chosen-Ciphertext Attack).

This means:

- attackers cannot distinguish or derive meaningful information,
- even if they have access to a decryption oracle.



Famous Example: Padding Oracle Attack

A classic example of CCA is the: Padding Oracle Attack against AES-CBC encryption.

In this attack:

- the system produces different responses for:
 - valid padding,
 - and invalid padding.

Simply by analyzing these error responses:

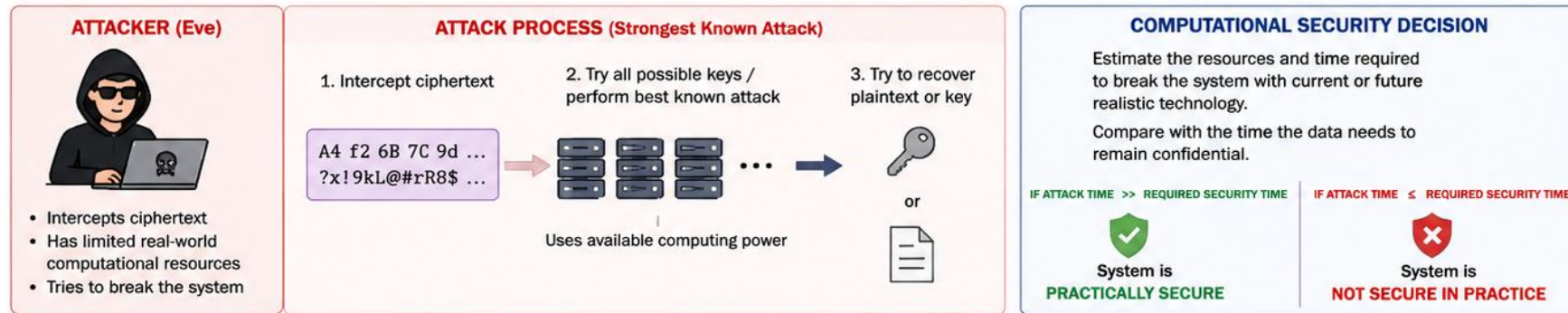
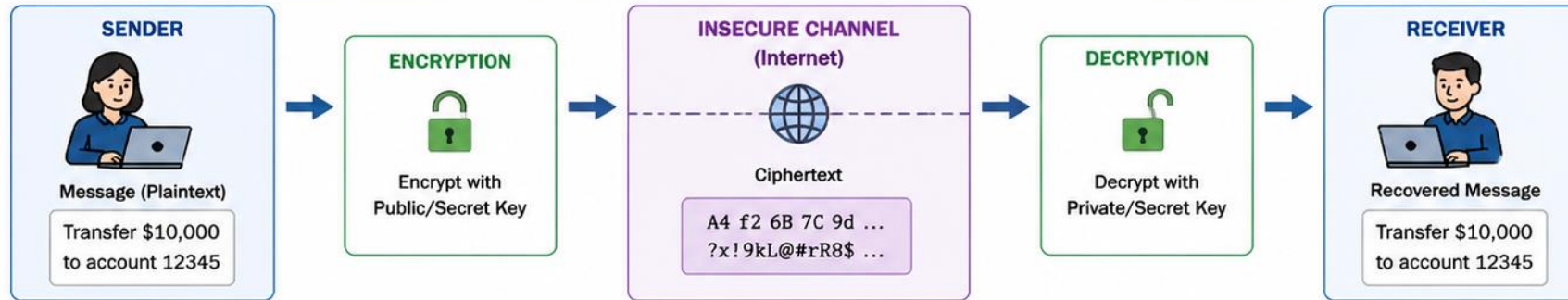
- the attacker can progressively decrypt ciphertexts without knowing the secret key.

This demonstrates that:

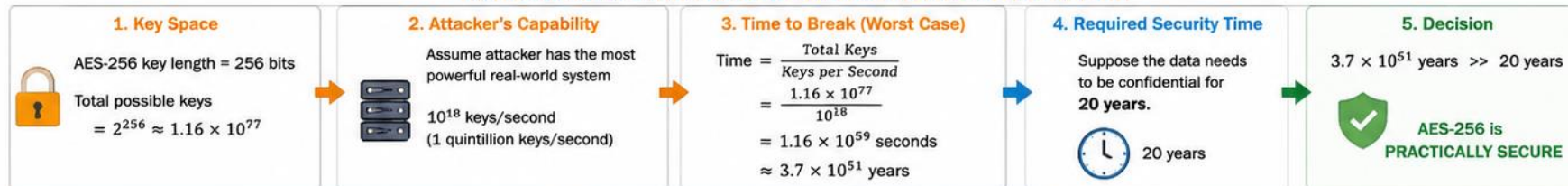
- even mathematically strong cryptographic algorithms may become vulnerable due to improper implementation.

COMPUTATIONAL SECURITY (PRACTICAL SECURITY)

A cryptosystem is secure if breaking it requires impractically large computational resources and time for any real-world attacker.



EXAMPLE: EVALUATING AES-256 AGAINST BRUTE-FORCE ATTACK



KEY POINTS

- ✓ Based on realistic assumptions about attacker's resources.
- ✓ Use the strongest known attack and best current technology.
- ✓ Estimate computation cost (time, memory, power).
- ✓ Compare with how long the data needs to be kept secret.
- ✓ Often based on hard mathematical problems (e.g., factoring, discrete log) even without a formal proof of equivalence.

COMMON HARD PROBLEMS USED

RSA	Integer Factorization Problem
ECC	Elliptic Curve Discrete Logarithm Problem
Diffie-Hellman	Discrete Logarithm Problem
Lattice-based Crypto	Lattice Problems (e.g., SVP, LWE)
....	

COMPARISON OF SECURITY LEVELS

Stronger
↑
↓
Weaker

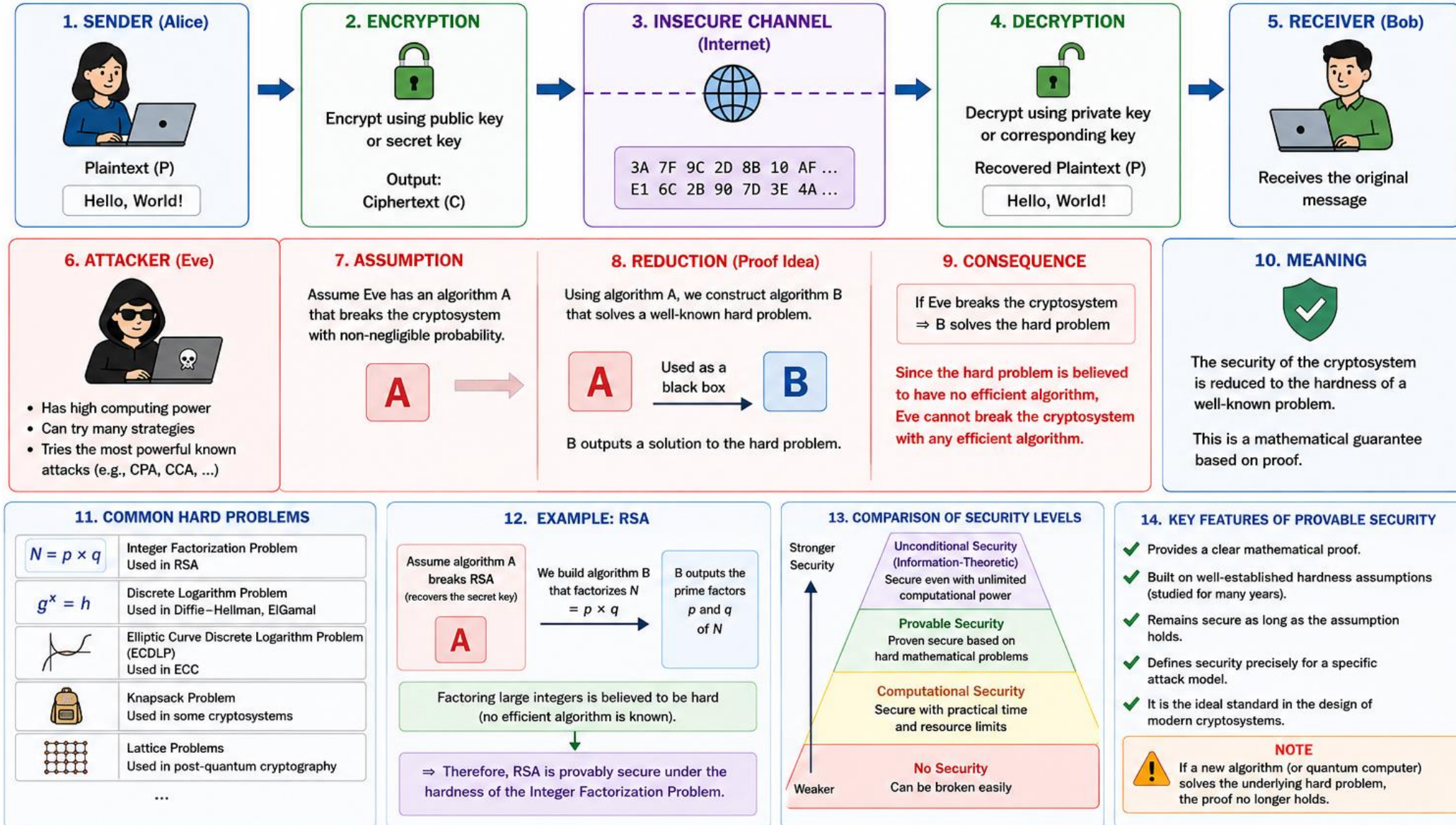
Unconditional Security (Information-Theoretic Security) Secure even with unlimited computational power
Computational Security (Practical Security) Secure within practical time and resource limits
No Security Can be broken easily



CONCLUSION: Computational security does NOT mean "impossible to break".
It means "the cost (time, money, resources) to break is greater than the value of the target or the time the data needs to stay secret".

PROVABLE SECURITY

A cryptosystem is provably secure if we can mathematically prove that breaking it is at least as hard as solving a well-known hard problem.



CONCLUSION

Provable security means: to break the cryptosystem, an attacker must solve a well-known hard problem, which is believed to be infeasible with any efficient algorithm.

UNCONDITIONAL SECURITY

Unconditional security represents the strongest security evaluation model in cryptography from the perspective of Information Theory. In this model, the security of a cryptographic system does not rely on any assumptions regarding the computational limitations of an adversary. The attacker is assumed to possess unlimited computational power, capable of exhaustively searching the entire key space or performing every conceivable cryptanalytic operation. Nevertheless, observing the ciphertext still provides no additional information about the plaintext.

A cryptosystem is said to achieve unconditional security if the probability distribution of the plaintext remains unchanged before and after observing the ciphertext. In other words, the ciphertext does not reduce the adversary's uncertainty about the plaintext. The ideal form of unconditional security is known as **Perfect Secrecy**, formally defined by Claude Shannon through the condition:

$$P(M = m \mid C = c) = P(M = m)$$

for every plaintext m and every ciphertext c with non-zero probability of occurrence. This condition implies that the ciphertext reveals absolutely no information about the plaintext. The classic example of a cryptosystem achieving perfect secrecy is the One-Time Pad, provided that the key is truly random, has a length equal to the plaintext, is used only once, and remains completely secret.

AD HOC SECURITY

Ad hoc security is a security evaluation model in which the security level of a cryptographic system is established primarily through the designer's own arguments, based on practical experience, empirical analysis, attack cost estimation, and resistance against certain known cryptanalytic methods. Unlike unconditional security or provable security, this model does not provide a formal proof that breaking the system is infeasible or computationally equivalent to solving a well-established hard mathematical problem.

Within this model, the adversary is generally assumed to be constrained by practical computational resources. The designer may argue that the best currently known attacks — such as brute-force key search, statistical cryptanalysis, differential cryptanalysis, or linear cryptanalysis — require computational efforts that exceed realistic feasibility. However, this type of security remains largely heuristic and depends heavily on the range of attacks considered during the evaluation process. Consequently, a cryptosystem achieving ad hoc security should not be interpreted as being formally secure; rather, it merely indicates that no practical attack method is currently known within the specific evaluation context.

INFORMATION HIDING TECHNIQUES

A BRIEF HISTORY OF INFORMATION HIDING

- The concept of information hiding dates back thousands of years and was primarily employed in military, diplomatic, and intelligence operations to protect confidential communications from enemy detection.
- However, only in recent decades has information hiding attracted significant attention from the research community and information technology institutes. The rapid advancement of digital technologies, large-scale data digitization, and the explosive growth of communication networks have been the major driving forces behind the development of this field.
- The ability to create nearly perfect digital copies, together with sophisticated techniques for modifying, manipulating, and distributing multimedia data over the Internet, has introduced numerous critical issues related to copyright infringement, unauthorized duplication, illegal distribution, and malicious content tampering. In this context, information hiding technologies have emerged as an important research area for supporting digital copyright protection, data authentication, integrity preservation, and enhancing information security within modern digital communication environments.



CONCEPT OF INFORMATION HIDING

Information Hiding is a technique of embedding digital information into another digital data object in such a way that the existence of the hidden information is difficult to detect.

Information hiding techniques generally pursue two primary objectives:

- protecting the confidentiality of the embedded data;
- protecting the carrier object itself.

These two objectives lead to two major branches of information hiding techniques:

- Steganography;
- Digital Watermarking.

Steganography

Steganography focuses on ensuring the secrecy and undetectability of the hidden information. Its primary objective is to conceal the existence of embedded data so that third parties cannot easily determine whether hidden information is present.

Steganographic techniques typically aim to:

- maximize embedding capacity;
- minimize detectability through statistical analysis or steganalysis;
- preserve the perceptual quality of the cover object after data embedding.

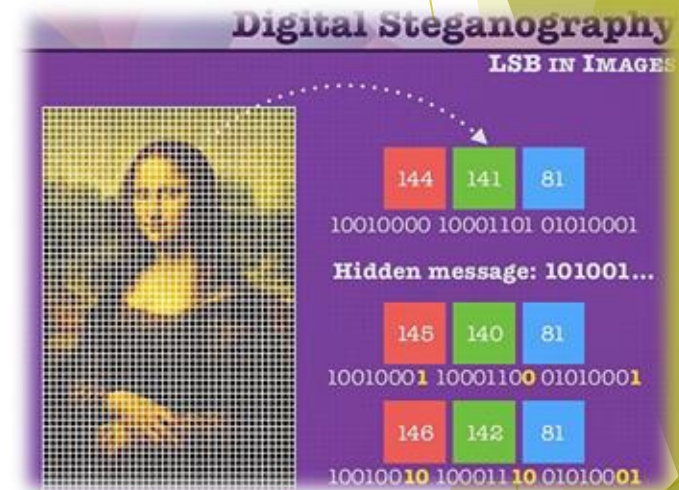
Digital Watermarking

Digital Watermarking focuses on protecting the carrier object itself rather than concealing the existence of embedded data.

Watermark information is embedded into digital media for purposes such as:

- source authentication;
- copyright ownership verification;
- tamper detection;
- integrity protection of digital content.

One of the most important requirements of digital watermarking is robustness, meaning that the embedded watermark should remain recoverable even after the host media undergoes compression, transformation, or moderate modification.



BASIC MODEL OF INFORMATION HIDING TECHNIQUES

To perform information hiding, it is necessary to design algorithms and embedding procedures capable of inserting hidden data into a cover medium. These procedures are responsible for embedding secret information into digital objects in such a way that the existence of the hidden data is difficult to detect while still satisfying specific security or protection requirements depending on the application.

In many information hiding systems, the embedding process employs a secret key similar to those used in cryptographic systems in order to enhance security and control access to the hidden information.

After the embedding process, a *stego object* is produced. This object can then be stored or transmitted through conventional communication channels without explicitly revealing the presence of embedded data.

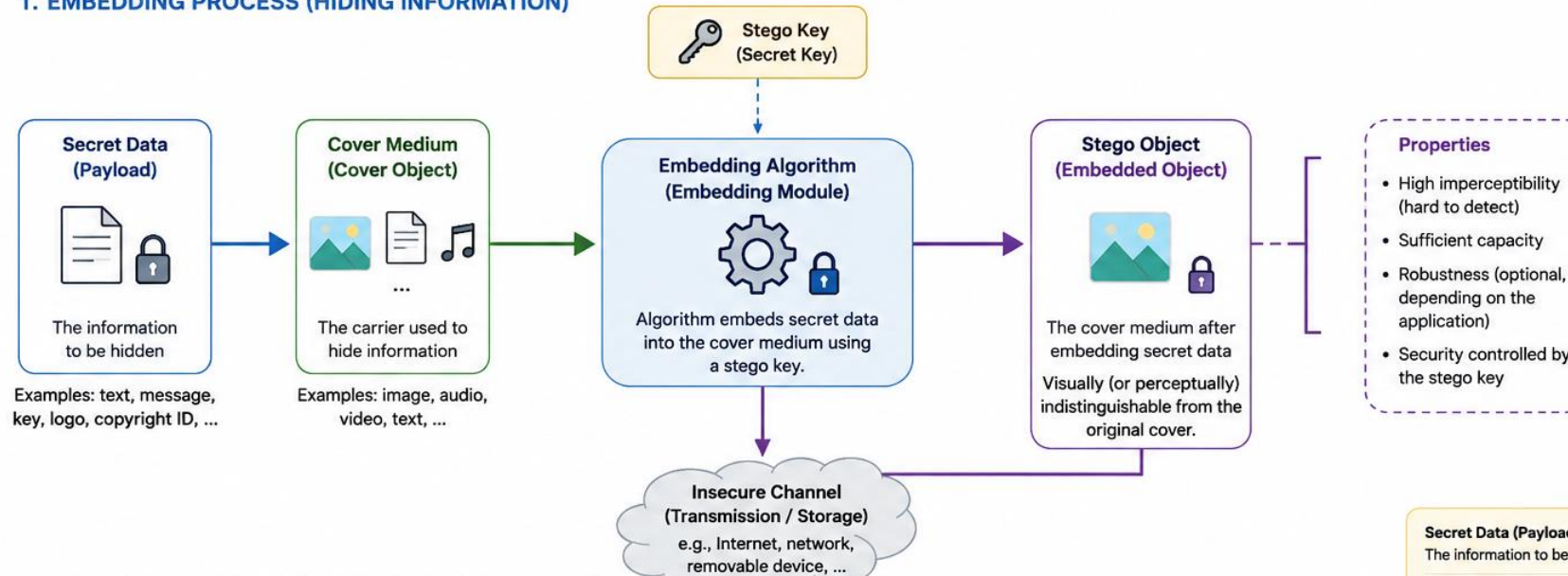
The processes of information embedding and information extraction are two inverse operations within an information hiding system and can be described through a model consisting of the following major components:

- **Secret Data / Payload:** The information to be embedded into the cover object. Depending on the application, this data may consist of confidential messages, cryptographic keys, logos, digital signatures, or copyright-related information.
- **Cover Medium / Cover Object:** The carrier used to conceal the hidden information, such as digital images, text files, audio, video, or other forms of multimedia data.
- **Embedding Algorithm / Embedding Module:** The algorithm or software component responsible for inserting hidden data into the cover medium according to a specific embedding method.
- **Stego Key:** A secret key used to control the embedding and extraction processes, thereby improving security and preventing unauthorized access.
- **Stego Object:** The resulting carrier object after the hidden information has been embedded.

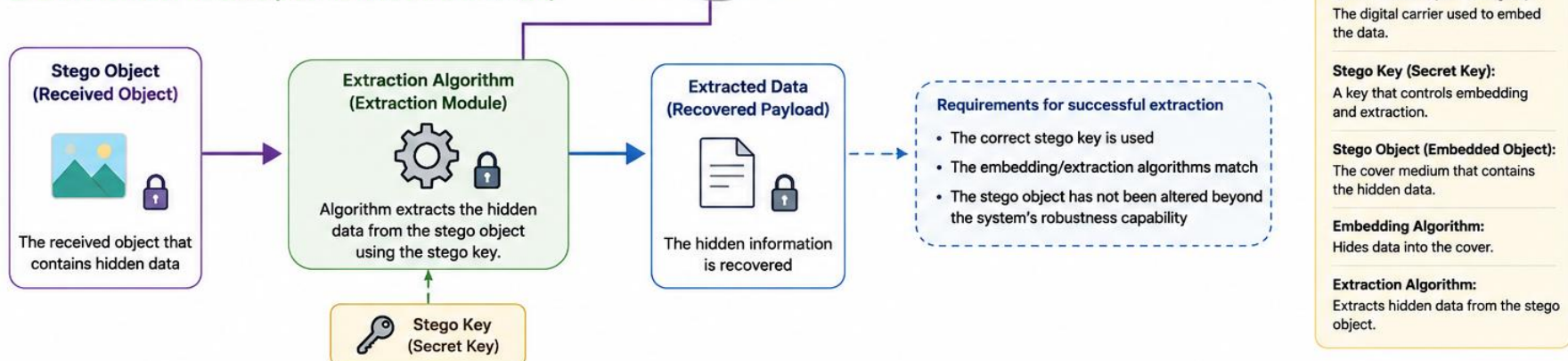
BASIC MODEL OF INFORMATION HIDING (STEGANOGRAPHY)

The embedding and extraction processes are inverse operations.

1. EMBEDDING PROCESS (HIDING INFORMATION)



2. EXTRACTION PROCESS (RETRIEVING INFORMATION)



INFORMATION HIDING MEDIA

Data Hiding in Digital Images

Data hiding in digital images is a major branch of information hiding in which digital images are used as the cover medium for embedding hidden data.

Currently, image steganography represents one of the most widely studied and deployed information hiding techniques due to the extensive use of digital images across the Internet and multimedia communication systems.

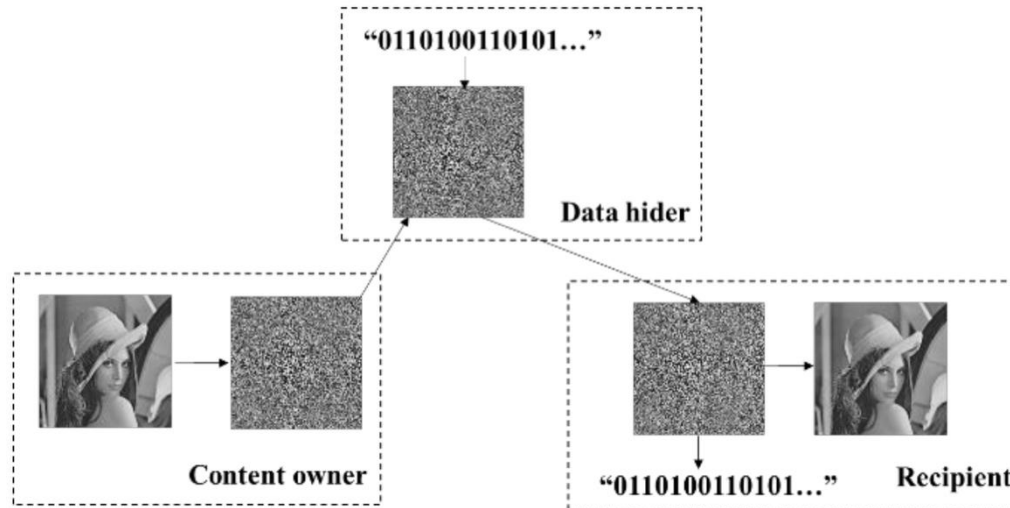
Image-based information hiding systems play an important role in various information security applications, including:

- data authentication;
- tamper detection;
- digital copyright protection;
- access control;
- secret communication.

As digital images become increasingly prevalent in modern society, image data hiding techniques have enabled numerous practical applications. For example, handwritten signatures may be digitized and embedded into electronic documents or banking authentication systems for identification and verification purposes.

In electronic identification systems such as digital identity cards, e-passports, and electronic credentials, authentication information may be embedded directly into facial images to support integrity verification and anti-counterfeiting mechanisms.

One of the most important characteristics of image steganography is imperceptibility. After the embedding process, the visual quality of the image remains nearly unchanged, especially for color and grayscale images, making the presence of hidden information difficult for human observers to detect.



INFORMATION HIDING MEDIA

Data Hiding in Audio

Audio data hiding possesses distinct characteristics compared with other multimedia hiding techniques because it relies heavily on the properties of the Human Auditory System (HAS).

A fundamental requirement of audio steganography is to ensure that the embedded information remains imperceptible while preserving the perceptual quality of the audio signal.

Whereas image steganography primarily exploits the characteristics of the Human Visual System (HVS), audio steganography leverages psychoacoustic properties of human hearing.

The human auditory system is highly sensitive to noise and capable of perceiving signals over a wide frequency range, making audio data hiding more challenging than image-based hiding techniques.

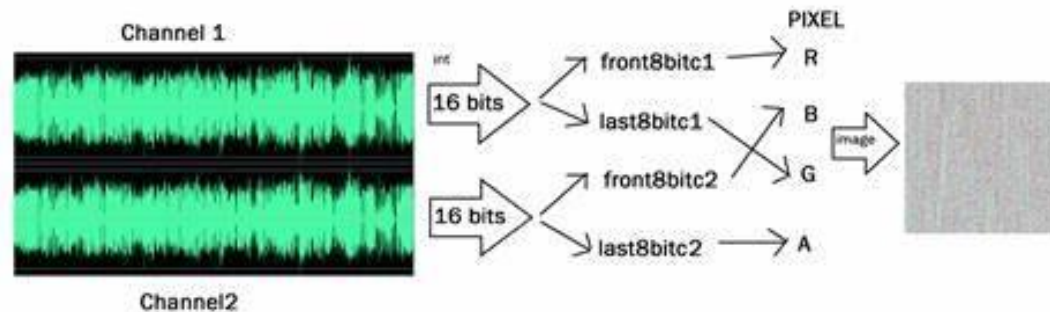
However, the phenomenon of auditory masking allows strong or high-frequency sounds to conceal weaker audio signals that are difficult for humans to perceive. This property forms the basis of many modern audio steganography techniques.

Psychoacoustic analysis models are commonly employed to identify suitable signal regions for data embedding.

In addition, audio data hiding is significantly affected by transmission conditions and channel bandwidth. Compression or transmission over low-quality communication channels may degrade or destroy the embedded data.

Consequently, audio steganography techniques typically require high levels of:

- synchronization;
- robustness;
- resistance against noise and signal processing operations.



INFORMATION HIDING MEDIA

Data Hiding in Video

Video data hiding is an extension of multimedia information hiding in which hidden data is embedded into video frame sequences or associated signal components.

Since video contains both visual and audio information, it provides significantly larger embedding capacity compared with single-image steganography.

Several video steganography approaches are based on spread spectrum techniques, where hidden data is distributed across multiple frequency regions or video frames in order to improve robustness and reduce detectability.

Many studies employ transform-domain methods such as:

- Discrete Cosine Transform (DCT);
- Discrete Wavelet Transform (DWT);
- Singular Value Decomposition (SVD);

to perform data embedding operations.

Modern video steganography techniques support not only image embedding but also the embedding of text, audio, and other multimedia data for applications including:

- copyright protection;
- content authentication;
- anti-piracy mechanisms;
- secret communication.

